

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

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Abstract

Long-horizon video generation is evaluated with whole-frame metrics that reward motion and temporal consistency. For fixed-camera nature scenes this creates an ambiguity: motion of water, fire, smoke, or rain is desirable, whereas motion of the background is an error. A system can therefore score well on motion while its scene drifts, or on consistency while its flow stagnates. We introduce SNF-Bench, an evaluation framework for long-horizon fixed-camera generation that partitions each scene into static support and dynamic flow and reports static fidelity, flow persistence with absolute magnitude, and drift leakage separately, never as one score. Each factor is validated mechanistically rather than by correlation with preference: we inject global translation, rotation, and scale drift and progressive motion attenuation at known severity into real generations, and require each factor to respond in its stated direction and to remain selective against corruptions it does not target. Auditing seven publicly released long-horizon generators in their intended configurations, we find that whole-frame motion and static-region drift induce near-opposite orderings of the same outputs, and that controlled drift which triples our static-fidelity factors moves a widely used whole-frame motion score by roughly a tenth—in the direction that rewards it. SNF-Bench measures where motion occurs and whether it persists; it does not measure physical realism.

1. Introduction

Autoregressive and streaming video diffusion models now extend generation from seconds to minute-scale horizons. Evaluation has not kept pace: small temporal errors invisible in short clips accumulate over long rollouts into background displacement, colour drift, or motion decay, and whole-frame metrics summarise these qualitatively different failures into one score.

The problem is sharpest in fixed-camera nature scenes. In a static-camera video of a river, the water should move while the bank, rocks, and surrounding landscape stay fixed. The same separation governs waterfalls, ocean waves, rainfall, fire, smoke, and wind-driven vegetation: some regions form static support that must remain structurally fixed, while others carry dynamic flow that must persist. Whole-frame motion measures reward optical flow regardless of its origin, crediting background drift as readily as genuine fluid motion, while temporal-consistency measures can favour videos whose stability comes from suppressing the flow they should sustain. Two opposite failures—static drift and flow decay—are thereby conflated, and increasingly so at long horizons, where accumulated background motion can dominate a sequence’s apparent dynamics. Figure 1 shows both failures in released generations of a single prompt.

We introduce SNF-Bench (Static-Natural Flow), an evaluation framework built around this decomposition, evaluated at 60 and 120s horizons with shorter and longer tiers retained as diagnostics. Measurement is organised into three factors. Static fidelity captures geometric displacement and residual optical flow inside nominally stationary regions. Flow persistence reports the retention of dynamic-region motion together with its absolute compensated magnitude, so a retention ratio cannot appear strong merely because motion is uniformly weak. Drift leakage relates static-region motion and global-motion compensation to the measured dynamic-region flow.

A benchmark is only as trustworthy as its measurements, so we validate the protocol independently of any model comparison. Controlled perturbations inject global translation, rotation, and scale drift, progressive motion attenuation, late freezing, photometric drift, and temporal repetition into real fixed-camera generations at known severity, and we require each factor to respond in the direction its definition states and to remain selective against corruptions it does not tar-



Figure 1. Why a fixed-camera benchmark needs a spatial decomposition. Three released 60s generations of one prompt. A whole-frame motion score rates the first two rows almost identically, yet only the first holds its scene in place; their static-region drift differs by over an order of magnitude. The third row shows the opposite failure, which that score does detect. DD* is the per-video quantity underlying Vbench Dynamic Degree, computed under its published rule but read before the binarisation the suite applies when aggregating, so that near-identical rows remain distinguishable as numbers.

get. Because a parameter expressed in degrees is not commensurate with one expressed in pixels, geometric severity is specified as the mean displacement a corruption induces, which makes the families directly comparable. We additionally recompute every measurement under systematic erosion and dilation of the region partition.

We then audit seven publicly released long-horizon generators in their intended configurations. The audit exposes cases where conventional metrics and SNF-Bench interpret the same outputs differently: high whole-frame motion coexisting with substantial static-region displacement, and high stability coexisting with decaying flow. We do not argue that existing metrics are incorrect. They answer a whole-frame question accurately; that question does not encode the spatial distinction this setting requires. Nor does SNF-Bench measure physical realism—a sequence may be drift-free and persistent yet physically implausible. The contribution is narrower and complementary: making the spatial origin and temporal persistence of motion explicit.

Our contributions are:

1. SNF-Bench, an evaluation framework for long-horizon fixed-camera generation organised around a static-support/dynamic-flow partition.
2. A measurement protocol that reports static fidelity, flow persistence with absolute magnitude, and drift leakage separately, and never as a single score.
3. Mechanistic validation of every reported factor under corruptions of known type and severity, with selectivity measured against the families each factor should ignore and every measurement recomputed under erosion and dilation of the region partition.
4. A public-model audit showing that this decomposition materially changes the interpretation and or-

dering of long-horizon behaviour relative to whole-frame metrics.

2. Related Work

Video-generation evaluation. General-purpose suites assess visual quality, temporal consistency, motion, and semantic alignment. Vbench [8] decomposes quality into dimensions including background consistency, motion smoothness and dynamic degree, and Vbench-2.0 [26] extends it toward intrinsic faithfulness; Eval-Crafter [13] learns a weighting of many metrics from user opinion; Video-Bench [4] scores generations with multimodal language models validated against human preference. Others specialise: T2V-CompBench [19] targets compositional binding, VMbench [11] perceptually aligned motion quality, and WorldScore [1] controllability and dynamics for world generation. These set the standards a benchmark paper must meet—explicit taxonomy, validated metrics, stated scale—but their measurements are whole-frame or sequence-level, and such an aggregate cannot reveal whether motion originates where intended or in displacement of nominally static structure. VMbench is the closest comparator, but it asks whether motion is perceptually plausible where we ask where it occurs and whether it survives compensation for global displacement. Our validation is correspondingly mechanistic—perturb by a known amount, require the expected response—complementing the human-alignment evidence those suites provide.

Long-horizon autoregressive generation. Autoregressive and streaming generators extend synthesis beyond short bidirectional clips by producing frames or temporal chunks causally, including CausVid [24], Self-

145 Forcing [7], Infinite-Forcing [9], Rolling-Forcing [12],
 146 Reward-Forcing [14], LongLive [22], and Causal-
 147 Forcing [27] with its few-step successor Causal-
 148 Forcing++ [25]; bidirectional systems such as LTX-
 149 Video [3] and Wan [21] provide short-horizon reference
 150 points. These systems differ substantially in temporal
 151 block size, training objective, memory and cache pol-
 152 icy, and horizon-extension mechanism [23], yet their
 153 evaluation typically relies on generic video metrics or
 154 method-specific qualitative analysis. That diversity
 155 motivates auditing each system in its released config-
 156 uration, and the evaluation gap motivates a system-
 157 agnostic question: does static support remain fixed
 158 while dynamic flow persists?

159 Image dynamics and nature-flow animation. Work
 160 on animating still images has long modelled natural
 161 motion explicitly: controllable fluid animation sepa-
 162 rates static image structure from specified fluid mo-
 163 tion [15], and Generative Image Dynamics models long-
 164 term scene motion through learned trajectory repre-
 165 sentations [10]. These works treat static support and
 166 dynamic content as distinct modelling targets; SNF-
 167 Bench elevates that distinction to an evaluation prin-
 168 ciple for long-horizon generators.

169 3. SNF-Bench Design

170 SNF-Bench evaluates long-horizon generation under a
 171 deliberately constrained setting: a fixed camera observ-
 172 ing a scene containing both stationary structure and
 173 persistent natural motion. This isolates two failure
 174 modes that are difficult to separate in unconstrained
 175 video—static drift, where nominally stationary struc-
 176 ture moves over time, and flow decay, where intended
 177 dynamics weaken or disappear. Every sequence is
 178 treated as a composition of two spatial roles, Ω_{static}
 179 for regions expected to remain geometrically station-
 180 ary and Ω_{flow} for regions expected to exhibit sustained
 181 motion. The partition defines an evaluation role, not a
 182 semantic ontology: a riverbank is static support while
 183 the river surface is flow; buildings are static support
 184 in a rainfall scene while the falling rain is flow. SNF-
 185 Bench does not require dynamics inside Ω_{flow} to follow
 186 any physical model. It measures whether motion stays
 187 localised to the appropriate regions and persists over
 188 time without contaminating static support.

189 Every system receives identical conditioning per
 190 item, and no prompt is rewritten for a particular
 191 method. Both conditioning modalities are evaluated
 192 under the same decomposition, differing only in how
 193 static support is established: the text-conditioned set-
 194 ting builds it in the opening of the sequence, while the
 195 image-conditioned setting takes it from a shared source

image, so divergence there is attributable to the sys-
 tem rather than to a differently imagined scene. The
 main text reports the text-conditioned track for space;
 complete image-conditioned results appear in the sup-
 plementary material.

3.1. Scene Categories and Horizons

Prompts are stratified across representative fixed-
 camera natural processes—channel water, ocean waves,
 precipitation, fire and smoke, lava, and wind-borne
 motion—with the category set fixed before the final
 sweep. The strata are not equally populated: chan-
 nel water accounts for 61% of the principal prompt
 set. Reported aggregates are means over prompts, not
 category macro-averages: three of the five populated
 categories hold two prompts each, so equal category
 weighting would buy balance at a cost in variance the
 sample cannot support. The supplementary material
 instead recomputes every system both ways and states
 which conclusions survive, with complete per-category
 counts.

Four temporal regimes are used: 5s as an initialisa-
 tion check, 60s as the reported horizon, and 120 and
 240s as diagnostic tiers. Conclusions rest on the 60s
 tier, which carries 23 prompts; the longer tiers are re-
 ported in the supplementary material, since their sam-
 ples do not support method-level separation on their
 own. A system that cannot reach a horizon under its re-
 leased configuration is reported as unsupported rather
 than silently modified.

3.2. Region Partition

The partition is obtained automatically and identically
 for every sequence. Mean optical-flow magnitude over
 the leading 12% of sampled frame pairs is thresholded
 by Otsu’s method [16] into a dynamic region and its
 complement; the static region is then eroded and a
 4% frame border discarded, which together form an ig-
 nored transition band so that boundary pixels—where
 the partition is least certain and flow least reliable—
 contribute to neither region.

Restricting the estimate to an early window prevents
 circularity: deriving a partition from the entire mo-
 tion field would let accumulated drift be read as evi-
 dence that the affected region is dynamic. At 60s the
 partition is fixed from the first 7.1s and the metrics
 evaluated over the remainder, so it reflects each sys-
 tem’s layout—unavoidable when every system authors
 its own scene—but not the temporal failure being mea-
 sured.

Two limitations bound the later conclusions. The
 partition is automatic, so a system drifting within the
 first 12% of its rollout can influence its own partition;

we therefore treat partitions as measured quantities and recompute every result under systematic erosion and dilation of the boundary (Sec. 5). And a two-way partition cannot represent layered content: rain and snow transit static support, so those pixels are assigned by flow magnitude alone and static-fidelity values in precipitation scenes carry an unmodelled contribution. Derivation, the sensitivity study, and the annotation protocol we regard as the natural strengthening of this component are in the supplementary material.

3.3. Benchmark Manifest

SNF-Bench is defined by a versioned manifest fixed before the final sweep, recording per item: identifier and category, exact prompt, target horizons, resolution and frame-rate metadata, and provenance. After freezing, items are never removed because a system performs poorly; corrections require a new benchmark version. A failure taxonomy—unsupported horizon, out-of-memory, runtime crash, numerical corruption, invalid output—is likewise fixed in advance, and failures are reported alongside successful generations rather than silently excluded. Coverage is counted from valid per-video measurements rather than from the presence of a results file, so a run that terminated early is reported as partial.

4. Drift–Motion Evaluation Protocol

SNF-Bench evaluates fixed-camera generation along three complementary factors: Static Fidelity, Flow Persistence, and Drift Leakage. No individual score is sufficient: low background motion can be achieved by freezing the entire video, and high foreground motion can be produced by drifting the entire scene. We therefore always report static-region stability jointly with both the relative persistence and absolute magnitude of dynamic-region motion.

Let I_t denote frame t of a T -frame sequence, Ω_{static} and Ω_{flow} the fixed region masks of Sec. 3.2, and $\mathbf{u}_t(\mathbf{x})$ the optical flow from I_t to I_{t+1} at pixel $\mathbf{x}$, estimated with RAFT [20]. Each sequence defines an early window $\mathcal{E}$ and a late window $\mathcal{L}$ of equal duration; $\mathcal{E}$ begins after a short warm-up interval so that initial generation transients do not define the motion reference. Window extents are fixed in the benchmark specification.

4.1. Static Fidelity

Feature-aligned Background Drift (fBD). We extract ORB features [18] inside Ω_{static} from the first frame and match them against each frame I_t in the late window, applying robust geometric filtering to reject outlier cor-

respondences. For the resulting matched pairs $\mathcal{M}_t$,

$$D_{\text{feat}}(t) = \text{median}_{(p,q) \in \mathcal{M}_t} \|p - q\|_2,$$

$$\text{fBD} = \frac{100}{d} \cdot \frac{1}{|\mathcal{L}|} \sum_{t \in \mathcal{L}} D_{\text{feat}}(t) \quad (\downarrow), \quad (1)$$

where d is the frame diagonal, so fBD is expressed as a percentage of image size. fBD captures accumulated geometric displacement of persistent structure and is intentionally insensitive to fine-scale flicker that moves no structure.

Normalized Background Flow (NBF). Complementing accumulated drift, NBF measures instantaneous motion energy inside the static mask:

$$\text{NBF} = \frac{10^3}{W \Delta t} \cdot \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{1}{|\Omega_{\text{static}}|} \sum_{\mathbf{x} \in \Omega_{\text{static}}} \|\mathbf{u}_t(\mathbf{x})\|_2 \quad (\downarrow), \quad (2)$$

normalized by frame width W for resolution invariance and by the inter-frame interval Δt for frame-rate invariance, giving units of 10^{-3} frame widths per second. Here Δt is the interval of the sampled pairs actually used for flow estimation: methods are evaluated at their native frame rates, which differ across the roster, and the evaluation pipeline subsamples each clip to a common temporal rate before measurement. Expressing NBF per unit time rather than per frame makes the quantity independent of both the native rate and the sampling policy, so the measure remains comparable if either changes.

NBF is a normalized magnitude, not a ratio: we deliberately do not divide by whole-frame flow, since that would couple background leakage to foreground behavior and inflate the measure for methods whose dynamic motion decays.

4.2. Flow Persistence

Global-motion compensation. Coherent scene displacement is removed before dynamic-region motion is evaluated. A single translation vector is insufficient: under rotation or zoom the static-region flow field is not constant, and its component-wise median is near zero for a centered rotation, so a translation-only estimate would leave exactly the corruptions Sec. 5 injects uncompensated. We therefore estimate a robust global similarity transform $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$ from static-region correspondences $\mathbf{x} \mapsto \mathbf{x} + \mathbf{u}_t(\mathbf{x})$ under RANSAC, define the induced global displacement field $\mathbf{d}_t(\mathbf{x}) = T_t(\mathbf{x}) - \mathbf{x}$, and compensate pointwise:

$$\tilde{\mathbf{u}}_t(\mathbf{x}) = \mathbf{u}_t(\mathbf{x}) - \mathbf{d}_t(\mathbf{x}). \quad (3)$$

338 We restrict the model to a similarity rather than an
 339 unrestricted affine transform because affine shear can
 340 absorb legitimate local deformation and over-correct.
 341 Where correspondences are insufficient or the robust
 342 fit fails we fall back to median translation and record
 343 that fallback per sequence.

344 Motion-Compensated Foreground Flow (MCFF). For
 345 a temporal window $\mathcal{W}$,

$$\text{MCFF}(\mathcal{W}) = \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} \frac{1}{|\Omega_{\text{flow}}|} \sum_{\mathbf{x} \in \Omega_{\text{flow}}} \|\tilde{\mathbf{u}}_t(\mathbf{x})\|_2 \quad (\uparrow). \quad (4)$$

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 347 We report $\text{MCFF}(\mathcal{E})$ and $\text{MCFF}(\mathcal{L})$, so a method
 348 with uniformly negligible motion cannot appear strong
 349 through relative measures alone.

Flow Persistence (FP).

$$\text{FP} = \min\left(\frac{\text{MCFF}(\mathcal{L})}{\text{MCFF}(\mathcal{E}) + \epsilon}, c\right) \quad (\uparrow), \quad (5)$$

351 clipped at a fixed constant c to prevent unstable ratios
 352 when early motion is small. FP near one indicates
 353 retained motion; substantially smaller values indicate
 354 decay; values above one are not automatically better.
 355 FP is never reported without its MCFF magnitudes.

356 4.3. Drift Leakage and Attenuation

357 Whole-frame metrics cannot determine whether appar-
 358 ent dynamic-region motion arises from intended flow or
 359 from displacement of the whole scene; we report two di-
 360 agnostics, since neither subsumes the other. Let F_{static}
 361 denote mean flow magnitude over Ω_{static} , and F_{raw} ,
 362 F_{comp} the mean raw and compensated magnitude over
 363 Ω_{flow} , all in the late window, where drift has accumu-
 364 lated.

Drift Leakage Ratio (DLR).

$$\text{DLR} = \frac{F_{\text{static}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow) \quad (6)$$

366 locates the sequence’s motion energy. It is not a frac-
 367 tion and is deliberately unbounded: $\text{DLR} > 1$ indi-
 368 cates that static support moves more than the region
 369 intended to move, a regime that occurs in practice.

Drift Attenuation Ratio (DAR).

$$\text{DAR} = 1 - \frac{F_{\text{comp}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow) \quad (7)$$

371 measures the signed relative change in dynamic-region
 372 flow after compensation—how much apparent motion

a global displacement model accounts for. Reported
 values are clipped to $[0, 1]$ while the signed quantity
 is retained for validation; negative values, which arise
 where local flow opposes the estimated global field, are
 reported rather than suppressed.

Interpretation. Neither quantity is a causal decompo-
 sition of motion into drift and non-drift components,
 and we do not describe them as one. Read together
 they are more informative than either alone: high DLR
 with moderate DAR indicates large static-region mo-
 tion energy that a rigid global model does not explain,
 implicating non-rigid instability such as warping rather
 than uniform scene translation.

Semantic guardrail. The decomposition is agnostic to
 what moves: a sequence rendering drifting fog where a
 river was requested can occupy a favourable operating
 point. The natural guardrail is a prompt-alignment
 score—CLIP-based image-text similarity [6], or a
 learned video-quality predictor [5] at higher cost. We
 do not report one, and the factors should be read
 accordingly: they establish where motion occurs and
 whether it persists, not that the depicted phenomenon
 is the one requested. Pairing SNF-Bench with a se-
 mantic score is how it should be used in practice.

5.4. Joint Interpretation

No factor is read alone: low drift with low compensated
 motion indicates an over-stabilised sequence, high com-
 pensated motion with high static-region flow indicates
 motion contaminated by displacement, and high reten-
 tion with negligible magnitude indicates persistent but
 trivial motion. SNF-Bench therefore reports no single
 scalar score, since collapsing the factors would recreate
 the ambiguity it exists to expose.

Implementation convention. Videos are normalised to
 a common evaluation resolution and subsampled to a
 common flow-estimation rate; no frame interpolation
 is performed, since synthesising frames would alter the
 flow field being measured. Every estimator, threshold,
 window extent and constant is fixed globally before the
 sweep and applied unchanged to every system, and the
 per-sequence frame rate, resolution and compensation
 mode are released alongside the scores.

5. Controlled Validation

Before SNF-Bench is used to compare generators, we
 validate that its measurements respond to the failure
 modes they target. A benchmark is not justified merely
 because its metrics rank differently; each must change

selectively, monotonically, and in the expected direction under known corruptions. We therefore construct controlled variants of fixed-camera reference clips with known corruption type and severity, independently of the audit in Sec. 6: perturbation strengths, masks, windows, and hyperparameters are fixed before the final results are inspected.

5.1. Perturbations and Criteria

Given a reference sequence we generate variants that isolate individual components of the decomposition. Geometric drift applies a temporally increasing global transformation—translation, rotation, or scale—simulating camera-like background displacement that accumulates as long-horizon error does. Motion attenuation progressively blends dynamic-region content toward an early reference, and late freezing replaces all content after a chosen fraction of the sequence, both leaving static support untouched. Photometric drift varies brightness and colour temperature while preserving geometry and motion, bounding the residual sensitivity of flow-based quantities. Temporal repetition replaces late dynamic content with a short repeated cycle. Partition perturbation erodes and dilates the region boundary rather than the video.

Geometric severity is specified as the mean pixel displacement a corruption induces, with each family’s parameter derived from it using the frame geometry. This is not presentational: a parameter in degrees is not commensurate with one in pixels, and at the evaluation resolution one degree of rotation induces only a few pixels of displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude.

A factor is admitted only if its response direction matches its definition, that response is monotonic in severity, unrelated corruptions produce substantially smaller changes, and conclusions survive perturbation of the region partition. Factors failing these checks are reported as diagnostics rather than headline quantities. The severity specification and the admission outcome for every factor are in the supplementary material.

Complementary checks. Mechanistic validation shows a factor responds to the corruption it names, not that its ordering matches what an observer perceives, and we do not close that gap here. The supplementary material specifies an axis-specific protocol—forced choice on one failure axis, never on which output is better—and reports the one instance we ran: an open-weight vision–language judge [17] failed a presentation-order control, so no agreement rate is drawn from it. Perceptual agreement remains open, and we claim nothing

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	drift	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	✓

Table 1. Mechanistic validation and factor admission. Each factor is scored on the corruption its definition says it should detect: accumulating global drift for the static-fidelity and drift diagnostics, progressive freezing for the persistence factors. ρ is Spearman correlation against injected severity, signed so that a correct response is positive; ρ_{rot} repeats it for rotational drift, the harder case. off-target is the largest relative excursion under corruptions the factor should ignore—photometric drift and perturbation of the region partition—so a small value indicates selectivity. Admission requires $\rho \geq 0.7$ and off-target response below 25%. Twelve reference clips per severity level.

about it.

5.2. Validation Results

Figure 2 and Table 1 report the outcome. Injected translation produces the expected monotone increase in both static-fidelity factors, and progressive attenuation of dynamic-region content drives MCFF-L and FP down while leaving the static factors comparatively unchanged—the selectivity the decomposition requires. The comparison that motivates the benchmark appears in the same experiment. Injected displacement that carries fBD to $3.2\times$ its unperturbed value moves VBench Dynamic Degree to only $1.1\times$, and in the wrong direction: a whole-frame motion score is largely insensitive to background drift, and to the extent it responds at all it improves on precisely the corruption a fixed-camera evaluation should penalise.

Two results qualify the framework rather than support it. Rotation is the harder geometric case for every factor, since correspondences degrade and the induced displacement is smallest at the frame centre: both static-fidelity factors respond more weakly than under translation (+0.70 for fBD, +0.60 for NBF, against +1.00 for both), so neither is used alone. DLR is the sharper limitation—admitted on its target family, where it responds perfectly, but with an inverted rotational response (−0.80), so it is read only as a comparative diagnostic alongside DAR. Reported alongside these is the scope boundary of Sec. 5.1: looping content sustains both persistence and magnitude, so SNF-Bench cannot distinguish genuine progression from repetition, and does not claim to.

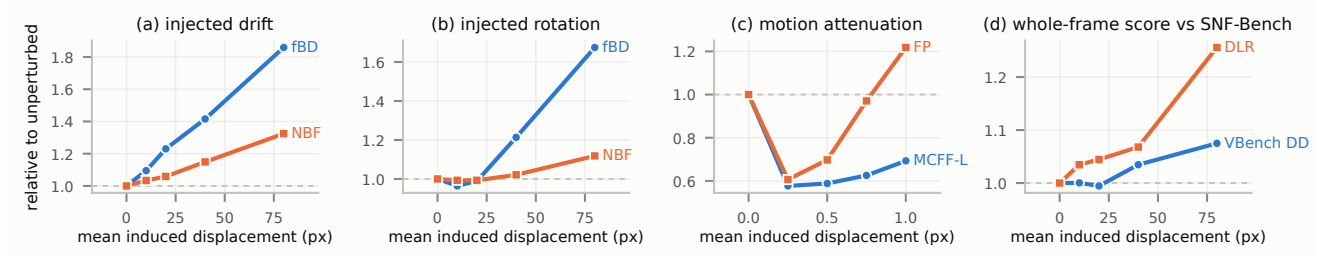


Figure 2. Mechanistic validation. Each panel injects a corruption of known kind and severity into real fixed-camera clips and plots each factor relative to its unperturbed value. Panel (d) puts VBench Dynamic Degree through the identical corruption as the SNF factors. Displacement that raises fBD to $3.2\times$ its unperturbed value moves the whole-frame score by $1.1\times$ —and upward: the generic score is close to blind to where the motion originates, and what little it registers it counts in the drifting system’s favour. The perturbation is synthetic, so the comparison implicates no published method.

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6. Public-Model Audit

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We apply SNF-Bench to publicly released long-horizon generators. The goal is not a universal leaderboard but a determination of how deployed systems occupy the static-fidelity/flow-persistence space when used as their authors intended.

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Systems and setting. The audit covers 7 publicly released autoregressive or streaming generators with accessible checkpoints and reproducible inference code, frozen as a set before the sweep. Unpublished and internal systems are excluded by construction, as are systems whose outputs cannot be independently reproduced. Each is run in the deployment regime its release intends: differences in temporal block size, denoising schedule and memory policy are preserved deliberately, being part of the released system rather than nuisance variables. The audited systems share a deployment envelope—all generate 832×480 at 16 fps and reach 240s without modification to their released temporal policy—so no comparison here is confounded by output format. We never substitute a block size, cache policy or schedule to populate a table entry; a system that cannot reach a horizon unmodified is marked unsupported.

The image-conditioned track is audited under the same protocol and reported in full in the supplementary material; there, systems evaluated under a common long-horizon wrapper rather than their released configuration are labelled as such, and their scores are read as deployment sensitivity rather than as the published method’s performance.

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Controls and statistics. Within each benchmark item every system receives identical conditioning, and the seed set is fixed before the sweep. Seed control guarantees reproducibility within a system; because imple-

mentations consume randomness differently, identical seeds do not induce matched noise draws across systems, so cross-system comparison rests on the paired-item design rather than on paired noise. Generated videos are retained in native form, and the normalisation of Sec. 4 is applied unchanged to every system.

Scores are computed per item and averaged over items; the sensitivity of every conclusion to category macro-averaging is reported in the supplementary material. Uncertainty is estimated by percentile bootstrap [2] over items, and we report 95% intervals. Exact ordinal ranks are avoided where uncertainty does not support them: systems whose intervals overlap substantially are not described as separated. Operational limits—unsupported horizons, failed generations, memory exhaustion, invalid outputs—are reported as rates against the frozen manifest and never removed after inspection.

To test whether the decomposition changes interpretation, the same outputs are additionally scored with established whole-frame metrics and we compare the orderings they induce. The focus is interpretation change, not a claim that existing metrics are incorrect.

7. Results and Interpretation Changes

Unless stated otherwise, comparisons refer to systems in their released configurations; deployment-sensitivity results are reported separately in the supplementary material and never redefine a system’s native behaviour.

7.1. Native-Capability Audit

Table 2 reports SNF-Bench measurements at the 60s horizon, the tier whose sample supports method-level separation; all four horizons appear in the supplementary material.

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 23$)				
CausVid	8.08 ± 3.15	9.27 ± 4.14	$2.87 \rightarrow 1.96$	0.67 ± 0.18
Self-Forcing	11.4 ± 3.63	12.7 ± 3.97	$10.4 \rightarrow 2.54$	0.44 ± 0.22
Infinite-Forcing	7.75 ± 3.84	3.65 ± 0.88	$1.17 \rightarrow 0.29$	0.63 ± 0.22
Rolling-Forcing	16.8 ± 3.69	11.2 ± 2.70	$3.18 \rightarrow 1.57$	0.75 ± 0.29
Reward-Forcing	14.8 ± 3.71	8.51 ± 2.79	$3.81 \rightarrow 0.80$	0.49 ± 0.18
LongLive	15.9 ± 2.89	13.9 ± 4.76	$6.46 \rightarrow 2.41$	0.75 ± 0.28
Causal-Forcing	22.4 ± 2.26	87.6 ± 18.6	$9.84 \rightarrow 7.21$	0.83 ± 0.31

Table 2. SNF-Bench audit, T2V track, native configuration. All systems are public external models run under their own intended configuration. Our own systems are excluded by construction. DAR is reported clipped to $[0, 1]$; signed values are released. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel. Values are means over prompts $\pm$ half a percentile bootstrap 95% interval (10k resamples, fixed seed). MCFF-E and MCFF-L are given together because FP is a ratio and is not interpretable without the magnitude it is taken over; intervals overlap for several systems, and only non-overlapping comparisons are described as separated. Aggregation sensitivity and the DLR/DAR diagnostics are reported in the supplementary material.

571 Motion persistence. Dynamic-region behavior is eval-
 572 uated at the joint operating point (MCFF($\mathcal{L}$), FP),
 573 never FP alone: sustained flow requires non-trivial late
 574 motion and high retention. Reporting MCFF-E beside
 575 MCFF-L makes the degenerate case visible—a system
 576 with negligible motion throughout cannot reach a high
 577 FP without also displaying a small MCFF-E.

578 Static-dynamic operating regimes. Figure 3 places
 579 each system in the static-drift-surviving-flow plane, ex-
 580 posing qualitatively different regimes—low-drift/low-
 581 motion, high-motion/high-drift, and low-drift/non-
 582 trivial-motion—without collapsing the axes into a
 583 scalar score. The two extremes are informative in op-
 584 posite directions. Infinite-Forcing has the lowest nor-
 585 malised background flow, and is the only system whose
 586 interval on that factor excludes every other, while also
 587 showing the least surviving late motion; on feature-
 588 based drift its interval overlaps the next-ranked system,
 589 so it is not the least-drifting system outright. Causal-
 590 Forcing sits alone at the opposite corner, separated on
 591 background flow by a wide margin.

592 7.2. Drift Leakage Changes the Interpretation of 593 Apparent Motion

594 The effect is systematic rather than anecdotal. Table 3
 595 correlates each generic whole-frame metric with each
 596 SNF-Bench factor over method-level means: Dynamic
 597 Degree tracks static-region flow at $\rho = +0.93$, so the
 598 systems a generic metric reads as most dynamic are,
 599 closely, those whose backgrounds move most. Causal-

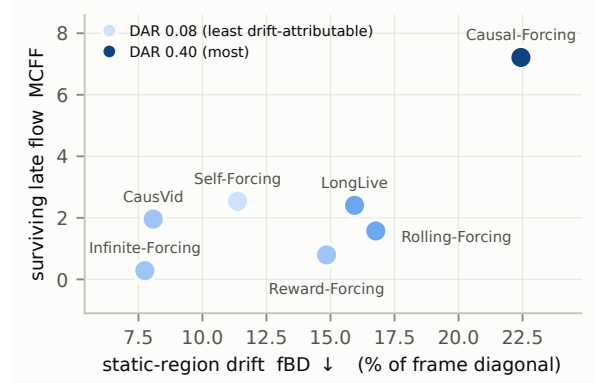


Figure 3. Operating regimes under native deployment. Colour encodes the relative attenuation of measured dynamic-region flow after global-motion compensation, not a share of it attributable to drift: the quantity is signed, and values are shown clipped to $[0, 1]$. The view exposes trade-offs; it is not a leaderboard.

generic metric	fBD	NBF	MCFF	DLR
VB-DD	+0.75	+0.93	+0.89	+0.36
VB-bg	-0.64	-0.71	-0.75	-0.50
VB-smooth / VB-flick	-0.96	-0.82	-0.61	-0.64

Table 3. Rank disagreement at 60s. Spearman correlation over $n = 7$ method means—descriptive, not inferential. Positive correlation between apparent motion and background flow is the effect the benchmark isolates. Smoothness and flickering rank these systems identically and share a row.

Forcing ranks 1 of 7 on Dynamic Degree and 7 on NBF; Infinite-Forcing mirrors it, ranking 1 on static fidelity and 7 on apparent motion. Read with its magnitudes, Causal-Forcing’s DLR of 0.94 indicates static-region motion energy almost equal to the region intended to move, while its DAR of 0.40 indicates a rigid global model accounts for only part of it—the remainder is non-rigid instability.

598 8. Limitations and Conclusion

SNF-Bench is intentionally narrow: long-horizon fixed-camera generation, where static support and dynamic flow separate. It measures neither physical realism nor semantic correctness, cannot distinguish progression from repetition, and rests on an automatic partition whose limits are stated in Sec. 3.2 and bounded by the erosion-dilation study. Conclusions unstable under that perturbation are never headline claims.

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